Question 1. Final project proposal

Name: _____ Student ID: _____ Group member (drone platform): _____

1. Topic

Learning a camera tracker from data: maximum-likelihood noise estimation and Bayesian decision rules for target tracking under fast vertical image motion, on a drone and an unmanned surface vessel (USV).

Background. Our group tracks one target with a camera on two platforms, using the same pipeline: YOLOv8 detection → Lucas-Kanade optical flow → Kalman filter. My teammate tracks a flying drone. I track the leader boat of a simulated USV fleet that keeps formation using only a camera. Both fail when the target's box moves up and down quickly in the image. The drone climbs and drops; the boat camera pitches in waves. On my USV logs, YOLO's missed detections rise from 1.8% to 14.7% as the box's vertical speed rises from 0–2 to 20–40 px per step.

Problem. The tracker's key parameters are hand-tuned for one platform: the process noise Q, the measurement noise R = 60·(0.5/confidence)², and a heuristic gate that decides which box is the target. Hand-tuned values do not transfer. On the boat, the drone tracker is worse than raw YOLO: 73% vs 85% of frames tracked with IoU ≥ 0.5 in clear water.

Approach (course methods).

1. **Maximum-likelihood estimation (Ch. 3-1).** R is the Gaussian MLE of the detection error, $\hat{\Sigma} = (1/n)\Sigma(e_k - \hat{\mu})(e_k - \hat{\mu})^t$. Q is the value that maximises the likelihood of the filter's innovations:

   $$\hat{Q} = \arg\max_Q \Sigma_k \ln N(v_k; 0, S_k(Q)), \quad S_k = HP_{k|k-1}H^t + R$$

   When ground truth exists, Q also has the textbook closed form $\hat{Q} = (1/n)\Sigma\, w_k w_k^t$, with $w_k = x_{k+1} - Fx_k$.
2. **Minimum-risk Bayes rule (Ch. 2)** to accept or reject each detected box. Features: confidence, Mahalanobis distance, size change. Accept if

   $$p(x|\omega_1)/p(x|\omega_2) > [(\lambda_{12} - \lambda_{22})/(\lambda_{21} - \lambda_{11})] \cdot P(\omega_2)/P(\omega_1)$$

   where accepting a false box costs more than rejecting a true one.
3. **PCA, Fisher LDA and the Hughes effect (Ch. 3-2, Ch. 1)** to choose and visualise the features, and a measured curve of accuracy against dimension and training-set size.

> **Preliminary result** (offline replay against ground truth; Q learned on training data only). Q chosen by maximum likelihood is 10× the hand-tuned value on the boat. That matches the frame rates: 10 Hz against 30 fps means about 3× the motion per step, so about 9× the variance. With it, the tracker is better than the hand-tuned one on **64/64 USV sequences** (median +13.7 points, Wilcoxon p = 4e-12). In clear water it reaches 88%, against 85% for YOLO alone. On the drone videos with a slow detector it gains +25 and +25 points (95% CIs exclude zero), and +24 points on a held-out video of a different drone. In pre-registered extreme waves (sea 4–5, heavy rain, dark storm) it still beats the hand-tuned tracker on **28/30 new sequences** (p = 3.5e-06). Against raw YOLO it ties or comes close in clear and rainy huge waves, but trails by about 10 points in the dark storm: that limit is what the final project will address.

2. Related papers

1. R. O. Duda, P. E. Hart, D. G. Stork, *Pattern Classification*, 2nd ed., Wiley, 2001 (Bayes decision theory, MLE, PCA, LDA).
2. R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Engineering*, 1960.
3. R. K. Mehra, "On the identification of variances and adaptive Kalman filtering," *IEEE Trans. Automatic Control*, 1970 (ML estimation of Q and R).
4. G. F. Hughes, "On the mean accuracy of statistical pattern recognizers," *IEEE Trans. Information Theory*, 1968.
5. R. A. Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, 1936.
6. Y. Bar-Shalom, E. Tse, "Tracking in a cluttered environment with probabilistic data association," *Automatica*, 1975.
7. B. D. Lucas, T. Kanade, "An iterative image registration technique with an application to stereo vision," IJCAI 1981.
8. A. Bewley et al., "Simple online and realtime tracking," ICIP 2016.

3. Project tasks and working schedule

Week	Task	Course link
Nov 5 – 11	Proposal and 3-minute talk. Datasets ready: USV runs with simulator ground truth; drone videos labelled by Grounding DINO and checked by eye.	Ch. 1 design cycle
Nov 12 – 18	MLE of R per axis and confidence; Q-Q plots to test the Gaussian assumption; closed-form Q from ground truth	Ch. 3-1
Nov 19 – 25	Q from innovation likelihood on both platforms; compare with the closed form	Ch. 3-1
Nov 26 – Dec 2	Minimum-risk Bayes gate; sweep $\lambda_{12}/\lambda_{21}$; ROC curves	Ch. 2, Ch. 1 (ROC)
Dec 3 – 9	PCA and Fisher LDA projection of candidate features; Hughes curve	Ch. 3-2, dimensionality
Dec 10 – 16	Full evaluation on both platforms; significance tests; side-by-side videos	Ch. 1
Dec 17 → final	Report, slides, rehearsal (dates adjusted to the announced final)	

4. Final product

1. A Python module shared by both platforms that replaces the hand-tuned constants: MLE noise model (R, Q), Bayes gate and feature analysis.
2. Labelled evaluation data: USV camera sequences with exact simulator ground truth, and drone videos with checked pseudo ground truth.
3. A reproducible offline replay that scores every tracker version against ground truth, with statistical tests.
4. Side-by-side detection videos (YOLO only | hand-tuned | learned) for both platforms.

5. Project report: components and items

Abstract · Introduction (fast vertical image motion on two platforms) · Related work · Problem formulation (state model, classes, loss matrix, minimum-risk rule) · Method (MLE derivations for R and Q, Bayes gate, PCA/LDA) · Data and ground truth · Experiments (tracking accuracy, fast-motion frames, ablation of each component, Hughes curve) · Statistical tests · Discussion and limitations (simulation only for the USV, pseudo ground truth for the drone, failure cases such as the dark storm) · Conclusion · References · Appendix (code).

6. Project presentation

About 15 minutes, both members: (1) two video clips of the failure, drone and boat, 2 min; (2) the shared pipeline and its hand-tuned parts, 2 min; (3) course methods: MLE derivation, minimum-risk rule, LDA, 3 min; (4) USV results: likelihood peak at 10×, accuracy tables and the 64/64 test, 3 min; (5) drone results and side-by-side video, 3 min; (6) what transfers between platforms, limitations, questions, 2 min.